

Consider the nature of past wars. The twentieth century was not the deadliest (in percentage of the total population), but it brought something new: the beginning of the Extremistan warfare—a small probability of a conflict degenerating into total decimation of the human race, a conflict from which nobody is safe anywhere.

A similar effect is taking place in economic life. I spoke about globalization in [Chapter 3](#); it is here, but it is not all for the good: it creates interlocking fragility, while reducing volatility and giving the appearance of stability. In other words it creates devastating Black Swans. We have never lived before under the threat of a global collapse. Financial institutions have been merging into a smaller number of very large banks. Almost all banks are now interrelated. So the financial ecology is swelling into gigantic, incestuous, bureaucratic banks (often Gaussianized in their risk measurement)—when one falls, they all fall.* The increased concentration among banks seems to have the effect of making financial crisis less likely, but when they happen they are more global in scale and hit us very hard. We have moved from a diversified ecology of small banks, with varied lending policies, to a more homogeneous framework of firms that all resemble one another. True, we now have fewer failures, but when they occur ... I shiver at the thought. I rephrase here: we will have fewer but more severe crises. The rarer the event, the less we know about its odds. It means that we know less and less about the possibility of a crisis.

And we have some idea how such a crisis would happen. A network is an assemblage of elements called nodes that are somehow connected to one another by a link; the world's airports constitute a network, as does the World Wide Web, as do social connections and electricity grids. There is a branch of research called “network theory” that studies the organization of such networks and the links between their nodes, with such researchers as Duncan Watts, Steven Strogatz, Albert-Laszlo Barabasi, and many more. They all understand Extremistan mathematics and the inadequacy of the Gaussian bell curve. They have uncovered the following property of networks: there is a concentration among a few nodes that serve as central connections. Networks have a natural tendency to organize themselves around an extremely concentrated architecture: a few nodes are extremely connected; others barely so. The distribution of these connections has a scalable structure of the kind we will discuss in [Chapters 15](#) and [16](#). Concentration of this kind is not limited to the Internet; it appears in social life (a small number of people are connected to others), in electricity grids, in communications networks. This seems to make networks more robust: random insults to most parts of the network will not be consequential since they are likely to hit a poorly connected spot. But it also makes networks more vulnerable to Black Swans.